Group 22 Team Document

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COMP 1712 Set F

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COMP 1712 – Set F – Group 22

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Team Working Standards (Lab 3)

- Shared OneDrive doc link (edit access for all + eram1@bcit.ca)
- https://bcit365-my.sharepoint.com/:w:/g/personal/yliu429_my_bcit_ca/EXcGAE4Hy7pOvYTMZ6APie4BLN5mGCBWc3sU0oUbr9V7ZA?e=heiKms
- Set F – Group 22
- Members:
    - Finn Wylie A01455692
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    - Austyn Chan A01401624
    - Fara Halterman A01480744
    - Kelsen Alloway A01435664

Working Standards covering:

Communication

- Primary communication channel: Discord> COMP 1712 team server
    - Message whenever, we will keep notifications off at night
- Response time targets: Weekdays ≤ 12h; Weekends/holidays ≤ 24h
- Tight schedules, therefore set tasks/deadlines ahead of time
- Communicate immediately if you need someone else to complete their work before you can continue
- If you are ever unsure, ASK

Decision-making

- If the decision will not affect others, make the decision by yourself
- If any change of direction, we need to discuss together
- “Good Enough is Good Enough”

Conflict resolution

- No such thing as a “Dumb Question”
- All conflicts are handled as a group
- If things escalate, we bring in a teacher
- If anyone raises a voice, we take a group time out

Meetings

- If in person meeting, plan ahead. No ad hoc meetings.
- Keep meeting minutes, Fara will do the minutes
- Meeting agenda beforehand.

AI & Academic Integrity

- Transparency and following class reqs

Team To Do List (Lab 3)

Item	Lead	Due Date
Stakeholder Table	Fara	end of lab
Initial Features / Requirements List	Austyn	end of lab
System Request	Faye	end of lab
Feasibility Analysis	Finn	end of lab
Benefits / Cost Table	Kelsyn	end of lab

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System Request (Lab 1)

System Request
Project sponsor: Ivor Merck – Managing Director, Target Enterprises (Pty) Ltd
Business Need: ● Un streamlined and physical work processes. ● Improve organization and accessibility of customer information. ● Data quality issues (duplicate customer records, inconsistent addresses, etc.) result in increased costs. ● Current paper-based processes slow order intake and status enquiries causing lots of manual work. ● Support scalability for expansion across South Africa and internationally. ● Improve efficiency in managing orders, distribution, and customer communications. ● Provide better tracking of customer tastes, preferences, and purchase history.
Business Requirements: **Customer Information Management** ● Maintain customer list and demographics. ● Capture and store order history, preferences, and shopping behavior. ● Group customers by preferences and predict future purchases. ● Ensure duplicate data is identified and removed. **Ordering & Inventory** ● Process and manage incoming and outgoing orders. ● Track distribution and dispatch details.

- Order and receive goods efficiently.

Automation & Integration

- Automate printing (labels, customer info for catalogues).
- Integrate POS, bookkeeping, payroll, and accounting systems.
- Produce performance and customer preference reports.
- Enable online access (customer-facing website / online store).

System Features

- Scalable and future-proof, supporting aggressive expansion.
- Secure access with proper clearance for sensitive data.
- Compatible with existing systems, while reducing reliance on outdated software.
- Easy for staff to use, with minimal hardware requirements.
- Support potentially multilingual interfaces.

Business Value: (Qualitative)

- Expand market reach from Cape Town to all of South Africa (and beyond). Increased staff efficiency and reduced manual errors.
- Reduced paperwork, postage, and catalogue duplication.
- Improved customer experience and satisfaction.
- Greater consistency and accuracy in orders and advertising.
- Better insights into customer and market trends.
- Cost savings through reduced labour, operating costs, and reliance on third-party payroll/accounting tools.
- Stronger integration across departments, leading to better cooperation and scalability.

Special Issues or Constraints:

- Staff training required before system go-live.
- Transition from mostly manual to digital processes will be significant.
- Current customer data requires restructuring.
- Existing hardware may not support modern systems.
- Payroll and accounting systems do not currently align with order systems.

- Implementation must prioritize ease of use and minimal disruption.

Stakeholder Table (Lab 1-2)

Stakeholder	Role/Org	Information Needs	Influence/Power	Interest	Preferred Communication Channel
Champion (Ivor Merck)	Target Enterprises Director	Overview of costs/benefits	High	High	MS Teams + Weekly Brief + meeting in person + emails
Marketing Manager (Michelle Li)	Target Enterprises Marketing	Customer list incl. addresses	Medium	High	Weekly Brief + Email (need to send a walkthrough and a clickable screen)
Finance Manager (Kate Merck-Vance)	Target Enterprises Finance	Cost of goods purchased; cost of goods out (shipping); cost of the project; payment batch reports	Medium	Medium	Weekly Brief
System Users	Target Enterprises Staff	Processing incoming orders; dispatch to customers	Low	High	Surveys, interviews, testing
Singapore org	Investor	Expansion/Financial Clarity	Low	High	Project sponsor as go between

Customers	User		Low	Med/High	Not directly; through marketing manager

Feasibility Analysis (Lab 2)

Technical (Can we build/integrate it?)

- The network is patchy, especially around dispatch, and shared PCs are causing congestion. We may need to upgrade both the network and add more machines. *(Fara & Finn – 4.2.6, Kelsen – pg.2 4.2, Faye & Austyn – 4.2.6)*
- A unified customer purchase history system is required. Currently, the lack of purchase visibility limits targeting and customer service. *(Faye & Austyn – 4.2.3, Kelsen – pg.2 4.2)*
- Manual paperwork and re-keying slow operations and lead to errors. A digital order system could improve flow and accuracy. *(Faye & Austyn – 4.4.1, Finn – 4.2.4, Kelsen – pg.2 4.2)*
- Digital carrier tracking and visibility into shipment status are needed to replace paper slips used by dispatch. *(Faye & Austyn – 4.3.2)*
- Operations teams have requested a single-screen view of dispatch details to improve usability. *(Fara – 4.4.4)*
- Integrating with the online order system is essential to reduce duplicate records and streamline processing. *(Faye & Austyn – 4.4.2)*

Economic (Should we build it?)

- Cleaning up duplicate records and normalizing addresses would reduce costs from returned mail, incorrect deliveries, and customer dissatisfaction. *(Fara & Finn – 4.3.3, Faye & Austyn – 4.2.2 & 4.4.2, Kelsen – pg.2 4.2)*
- Moving from manual to digital workflows will save time and reduce error rates in order processing. *(Kelsen's – pg.2 4.2, Faye & Austyn – 4.2.5)*
- Improving tracking and ensuring accurate order fulfillment will improve customer experience. *(Finn – 4.2.5, Faye & Austyn – 4.2.2)*
- Training is a necessary economic cost. It's essential to avoid user frustration and ensure smooth adoption of the new system. *(Fara's – 4.3.1 & 4.3.4, Kelsen's – pg.1 4.1)*

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Organizational (Will they use/support it?)

- The marketing manager is highly engaged and wants to see a working prototype early to provide feedback. *(Fara, Finn, Faye & Austyn – 4.4.2)*
- Leadership expects weekly checkpoints and visible progress (e.g., screenshots or working demos) to show stakeholders. *(Fara – 4.4.6, Kelsen's – pg.2 4.3)*
- The marketing team sees address cleanup and duplicate prevention as a key early win. They're invested in a successful Phase 1. *(Kelsen's – pg.2 4.3, Faye & Austyn – 4.3.4)*
- Changes must not disrupt critical financial and dispatch workflows, especially around month-end or during reconciliation periods. Some artifacts (e.g., external dispatch slips) must remain paper-based for now. *(Faye & Austyn – 4.3.7, Kelsen's – pg.3 4.4)*
- Purchasing needs stock movement visibility, and finance is open to learning the system early. *(Fara – 4.3.5, Finn – 4.4.1)*
- Dispatch staff will need to move from paper slips to digital carrier records to gain visibility into shipments. *(Faye & Austyn – 4.3.2)*
- There is limited but growing interest in using data for marketing and targeting, pending better system support. *(Finn – 4.2.3)*
- Sponsor and operations want the official order form to remain unchanged in the initial rollout. *(Fara – 4.4.6)*
- We must retain the current order intake artifacts and dispatch paperwork formats in Phase 1 to avoid confusion and maintain staff efficiency. These familiar tools are critical during peak operations. *(Fara – 4.1.1 & 4.1.3, Finn – 4.4.5, Kelsen – pg.1 4.1)*
- Any transition to digital systems will require training, particularly to ensure clean data entry and effective use by order clerks and marketing. *(Fara – 4.3.1 & 4.3.4, Kelsen – pg.1 4.1)*
- Legacy processes and tools should not be altered too soon; staff rely on familiar systems to perform efficiently. *(Finn – 6.3)*

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Benefits & Costs Table (Lab 2)

Type	Consolidated Description	Source(s)	Impact	Confidence	Contributors
Tangible Benefit	Reduced costs and errors from duplicate mailings (returns, postage, inconsistent addresses).	“We’re sending orders to the same person twice…” — Michelle (4.2); “Duplicate sends and returns are a waste.”; (4.2, b2)	H–L	2–5	Kelsen, Fara, Finn, Faye, Austyn
Tangible Benefit	Reduced employee hours on paperwork / manual re-entry (finance + order clerks).	“Paper and re-keying slow the flow and create errors.” — Ivor (4.2, b1); “I’m manually copying numbers…”	H–M	4–5	Kelsen, Finn, Faye, Austyn
Tangible Benefit	Increased marketing efficiency (clean mailing list, unified purchase history).	“Stop new duplicates and normalize addresses.” — Michelle (4.3); “No unified purchase history limits targeting…”	M–H	3–4	Fara, Finn
Tangible Benefit	Reduces errors in invoicing and results in faster, more consistent payments.	“Audit needs a clear trace from each order line to the totals.” — Kate (4.3)	H	4	Fara
Tangible Benefit	Increased efficiency in operations (faster responses, clearer order visibility).	“People ask, ‘Where is it?’ and we have to flip through notes…” — Jeff (4.2, b5)	H–M	3–4	Kelsen, Fara, Finn, Faye, Austyn
Intangible Benefit	Improved staff morale, well-being, and reduced stress (faster order entry, less searching).	“People ask, ‘Where is it?’…” — Jeff (4.2, b5); order clerks rely on forms (4.3); staff well-being (4.2, b6)	M	2–4	Kelsen, Fara, Finn, Faye, Austyn
Intangible Benefit	Better customer experience and satisfaction (fewer errors, less frustration).	“It also frustrates customers and makes us appear unprofessional.” — Michelle (4.2, b2)	H–M	2–3	Kelsen, Finn, Faye, Austyn
Intangible Benefit	Board confidence and scalability through early prototypes and previews.	“If you could show me something by week three or four…” — Ivor (4.3); Sponsor/Marketing previews (4.4)	M–H	3–4	Kelsen, Fara
Dev Cost	New hardware / network upgrades (shared PCs, patchy network).	“The network is patchy by dispatch…” — Jeff (4.2, b6)	M–H	3–4	Kelsen, Fara, Faye, Austyn
Dev Cost	Regular prototypes and quick updates (needed for board/sponsor).	“Short weekly checkpoints…”; “Show me something by week three or four…” — Ivor (4.3)	H	4	Fara, Finn
Dev Cost	Initial employee training (adoption risk if familiar workflows change).	“Marketing maintains the mailing list…” (4.3); “Adoption risk: altering familiar artifacts too soon.”	M	3–4	Kelsen, Finn

Dev Cost	Define fields, totals, and test cases for reports.	(4.4, b1)	H	3	Faye, Austyn
Ops Cost	Ongoing support, integration, and scalability (APIs, analytics, multi-site rollout, advanced reporting).	“Label automation and carrier APIs…” (5)	L–H	3–4	Kelsen, Finn
Ops Cost	System maintenance (general assumption + later phase).	Assumed; (4.4, b4)	L–H	2–4	Kelsen, Faye, Austyn
Ops Cost	Adding purchase history / dispatch details (time-consuming, may need extra staff).	“We can’t see what a customer bought last time.” — Michelle (4.2); courier dispatch notes (5, b1)	L–M	2–3	Fara, Faye, Austyn
Ops Cost	Stock movement tracking (warehouse/logistics staff time).	(5, b1)	M	2	Faye, Austyn
Ops Cost	Short-term manual form entry (don’t redesign in phase 1).	“Do not redesign the form in the first phase.”	L	4	Finn

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System Features & Requirements (Lab 1)

- The system shall maintain a customer list without duplicates
- The system shall warn when a new entry with duplicate customer information is created
- The system shall for order enquires should load in under 2 seconds on older mobile devices
- The system shall display tracking information on order enquiry screen
- The system shall never show full card numbers to support
- The system shall record carrier, service level, tracking number, ship date, and weight for each order
- The system shall generate financial totals by date and payment method both daily and weekly
- The system shall not update on Fridays or at the month end
- The system shall backup daily and be capable of rolling back to anything from the last 30 days
- The system shall keep an audit trail on who changed what and when
- The system shall be restrictive on who can access what parts (support can't touch payment data)
- The system shall always capture customer email and phone numbers
- The system shall keep the store delivery addresses in consistent format
- The system shall generate reports on stock movement (items in/out)
- The system shall generate reports on sales summaries (orders and revenue over a date range)

Later phases:

- handle incoming customer orders and dispatching them
- Organize distribution information
- Order new stock and receive goods from suppliers
- Stock/ordering management
- Link to finance/POS system
- Accounting & payroll system
- Budgeting, and allow correspondence
- Easily accessible

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Part 3 – User Stories

Epic: "System must provide a centralized Customer Database that tracks preferences, purchase history, and eliminates duplicate records."

Why is this one?

- It touches on customer experience, efficiency, and reporting.
- Everyone mentioned it in our group.
- It's broad enough to split into multiple user stories.

User Story 1

As a Marketing Manager, I want to see customer purchase history and preferences so that I can create targeted campaigns.

Acceptance Criteria:

- System shows purchase history per customer.
- Preferences (e.g., product categories) are stored and editable.
- Include a dashboard with export options for easy data access and visualization.
- Allow exports in specific file formats (e.g., CSV and Excel).
- Allow filters such as date range, product category, or customer segment.

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User Story 2

As a Customer Service Representative, I want to quickly look up a customer's order details so that I can respond accurately and efficiently to customer inquiries.

Acceptance Criteria:

- CSR can search for customers by company name, name, email, or customer ID.
- Customer record displays full order history.
- Latest order status (pending, shipped, delivered) is clearly shown.
- CSR has read-only access; sensitive data is masked.
- The system logs CSR access to customer records.

User Story 3

As a Data Entry Clerk, I want duplicate customer entries flagged and merged so that we don't waste time and postage on multiple records.

Acceptance Criteria:

- System automatically detects likely duplicate entries.
- Users can merge records without losing history.
- Confirmation required before deleting/merging.
- Warns if a new entry contains information already in the system.
- Audit and change logs available for merged information

User Story 4

As a Finance Officer, I want customer data integrated with the POS and finance systems so that I can reconcile payments with orders more easily.

Acceptance Criteria:

- Credit card numbers are hidden unless viewed by an account with certain permissions.
- Personal data is hidden or obfuscated on publicly viewable displays.
- Customer IDs sync with the finance system.
- Each transaction is linked back to a customer's record.
- Reports show sales by customers and by period.

User Story 5

As a Secretary, I want customer information easily available to make customer correspondences easier.

Acceptance Criteria:

- Search function by name, email, or customer ID.
- System shows purchase history per customer.
- System integrates to mail/other communication systems
- Automatic formatting of communication documents with customer information
- Customer information available in multiple formats (i.e. Envelope address, Email format, etc.)
- Unnecessary confidential information is redacted or hidden

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Team Retrospective (Lab 4)

What helped our coordination/collaboration/communication?

- Team members felt comfortable asking questions, giving feedback, and raising concerns early. (Faye)
- Having shared objectives and timelines gave structure and focus. (Faye)
- Clear division of task (Austyn)
- Open communication and collaborative language (Kelsen)

What hindered it?

- Lack of caffeine (Kelsen)
- Friday's being really really long (Kelsen)
- Staying on schedule for each task to finish when we want to
- People having different working speeds/initiation costs
- Some members contribute more while others less created imbalance. (Faye)
- Rushed deadlines sometimes reduced the quality of collaboration. (Faye)

Action for next lab:

- Better develop and document our decision-making process, ensuring everyone is involved.
- Clear goal/deadline

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Requirements Elicitation Strategy (Lab 4)

In order to triangulate project requirement elicitation, our team has chosen to combine structured interviews with key stakeholders to gather qualitative insights, direct observations of staff performing tasks to capture real-world workflows and challenges, and targeted surveys with scaled response options to quantify user preferences and measure satisfaction.

Data Source (Who)	Method (How)	Data Analysis Strategies
All managers	Interview	Human
All staff/users	Survey	Automatic analysis with the survey system
Variety of positions, primarily end user	Observational	Report

We have chosen this strategy because:

- A survey is an efficient way to gather broad feedback across multiple staff members, especially frontline workers.
- Interviewing managers will give us an overview of system requirements from different departments
- Interviewing managers instead of system users will reduce the time and resources needed to conduct interviews
- Observational provides real-world context by watching staff use existing systems and workflows in live settings.
- Observation can confirm whether reported issues found in survey and interview actually appear in practice.

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Kelsen:

Interview Questions (open-ended)

- Support team: “When you look up an order on your phone, what information do you need most quickly, and what frustrates you about current mobile screens? Also, what type of phone and operating system do you use when accessing the system”
- Warehouse team: “How do you currently record and track shipment details, and what challenges do you face when linking that information back to orders?”
- Finance team: “When reviewing payment batches, what checks or comparisons do you typically perform to confirm totals match the processor?”

Observational Requirements / Tasks (things to measure or watch)

- Mobile performance test: Time how long it takes the Order Enquiry screen to load on an older Android phone (target: ~2 seconds).
- Data entry workflow: Watch warehouse staff input shipment details (carrier, tracking, weight) and note any errors, delays, or unclear fields.
- Report accuracy check: Compare Finance’s generated Payment Batch Report totals with processor records to confirm they always match.

Survey Questions (with options or scales)

- Support staff mobile usability: “How easy is it to find the right order information on your phone?”
  Very difficult | Difficult | Neutral | Easy | Very easy
- Warehouse staff data capture: “How accurate and efficient is the current process for recording shipment details?”
  Very inaccurate/slow | Somewhat inaccurate/slow | Neutral | Accurate/efficient | Very accurate/efficient
- Finance report reliability: “How confident are you that the Payment Batch Report provides the correct totals every time?”
  Not confident at all | Slightly confident | Moderately confident | Very confident | Completely confident

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Fara:

Interview Questions

- How can the current ordering process be improved?

Observational Requirements (things to measure or watch)

- Order clerks: How long does it take them to find a document they need? For example, does it take them more time to sort through a file folder of customer information, or to get the documentation ready to process an order?

Survey Questions (with options or scales)

- Would your current role benefit from switching tasks from a physical paper environment, to a software solution? (Completely Disagree, Slightly Disagree, Unsure, Slightly Agree, Completely Agree)

Austyn:

Interview Questions (open-ended)

- What are the biggest challenges you face in managing your team's daily operations when dispatching orders?

Survey Questions

- How often do you experience technical issues? (Options: never, rarely, sometimes, often, very often)

Observation

- How long does it take staff to find a customer's order details?

Faye:

Observation:

- Observe how dispatch info is recorded (paper, clipboard, spreadsheet). (Warehouse)

Survey

- How often do you have to call or message another department to get tracking info? (Support)

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Interview

- How do you ensure that addresses, emails, and phone numbers are entered in a consistent format? For example, are there specific rules or standardized fields for postal codes, phone numbers, or email addresses? What challenges do you face in maintaining this consistency and overall data quality?

Finn:

Observation:

- Analyze how customer support staff interacts with the system - such as responding to an order inquiry or setting up a new customer and identify what information they access most and where they encounter difficulties. (support)

Survey

- What is your mobile device brand and rough age of the device (All)

Document mining

- What format is the addresses of incoming and outgoing orders in?

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Requirements Refresh + Categorization (Lab 4)

ID	Stakeholder	Requirement (one idea)	Category	Sub-Category	Phase
R-01	Support	The system shall display tracking info on the Order Enquiry screen or show “Not dispatched yet.”	Functional	Process	1
R-02	Support	The system shall restrict Support from viewing full card numbers (only last 4 digits).	Non-Functional	Security	1
R-03	Support	The system shall complete loading of the Order Enquiry screen on an older Android phone in under 2 seconds.	Non-Functional	Performance	1
R-04	Warehouse	The system shall allow Warehouse staff to record: carrier, service level, tracking, ship date, and weight linked to orders.	Functional	Process	1
R-05	Warehouse	The system shall produce a Stock Movement report (items in/out).	Functional	Process	1
R-06	Finance	The system shall generate a Payment Batch Report with totals matching the processor.	Functional	Process	1
R-07	Finance	The system shall prevent software releases on Fridays and at month end.	Non-Functional	Cultural/Policy	1
R-08	Finance	The system shall enforce role-based permissions so Finance can run reports.	Non-Functional	Security	1
R-09	Admin / Ops	The system shall provide daily backups and restore capability for 30 days.	Non-Functional	Operational	1
R-10	Admin / Ops	The system shall maintain a simple audit trail of who changed what and when.	Non-Functional	Security	1
R-11	Admin / Ops	The system shall enforce role-based permissions (Warehouse = dispatch updates, Support = no payment access).	Non-Functional	Security	1
R-12	Data Quality	The system shall capture customer emails and phone #s on the order form.	Functional	Information	1
R-13	Data Quality	The system shall warn about likely duplicate customers during entry.	Functional	Process	1
R-14	Data Quality	The system shall store delivery addresses in a consistent format.	Functional	Information	1
R-15	Management	The system shall produce a Sales Summary report (orders/revenue over date range).	Functional	Information	1

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Team Retrospective (Lab 5)

Part 1

What helped our coordination/collaboration/communication?

- Our ability to split up work has been great for our collaboration. We trust everyone to do their part and then review everyone's work together.

What hindered it?

- A variety of motivation levels.

Action for next lab:

- Focus on completing the lab *together and* aim to finish it in the lab. If it takes longer than lab time, stay at school to finish the lab on the same day.

Part 2

Team Member	Use Case
Kelsen	Record shipment details
Fara - R-12	Capture email and phone
Austyn – UC-10	Sales Summary report
Faye – UC-4	Export Customer Data (Allows the Marketing Manager to export a filtered set of customer purchase history and preferences for analysis or reporting purposes.)
Finn -- UC-14	The system shall store delivery addresses in a consistent format.

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Use Case, Data Flow Diagrams & Entity Relationship Diagrams (Lab 5-7)

”The system shall allow Warehouse staff to record: carrier, service level, tracking, ship date, and weight linked to orders.”

Use Case Name: Record Shipment Details (Kelsen)

ID: UC-6
Priority: Medium
Actor: Warehouse Staff

Description:
The Warehouse Staff records shipment details for confirmed customer orders, including carrier, service level, tracking number, ship date, and package weight. This ensures that shipping information is linked to the correct order for customer tracking and logistics management.

Trigger:
Warehouse Staff selects a confirmed order to fulfill and chooses the “Record Shipment” option.

Preconditions:

- Order exists in system with status = “Confirmed/Authorized.”
- The order has assigned items ready for shipment.
- Warehouse Staff is authenticated and authorized to update shipping information.

Normal Course:

1. System displays list of confirmed orders pending shipment.
2. Warehouse Staff selects an order to record shipment details.
3. System displays shipment detail form showing order summary (items, buyer info, address).
4. Warehouse Staff enters:
    a. Carrier (e.g., UPS, Canada Post, FedEx)
    b. Service Level (e.g., Standard, Express, Overnight)
    c. Tracking Number
    d. Ship Date
    e. Weight (per parcel or total shipment)

5. System validates all required fields and data formats (e.g., tracking number pattern, date format, weight numeric).
6. Warehouse Staff submits the shipment record.
7. System saves shipment details and links them to the order record.
8. System updates the order status to “Shipped.”
9. System sends shipment confirmation (email + on-screen) to the Buyer with tracking info.

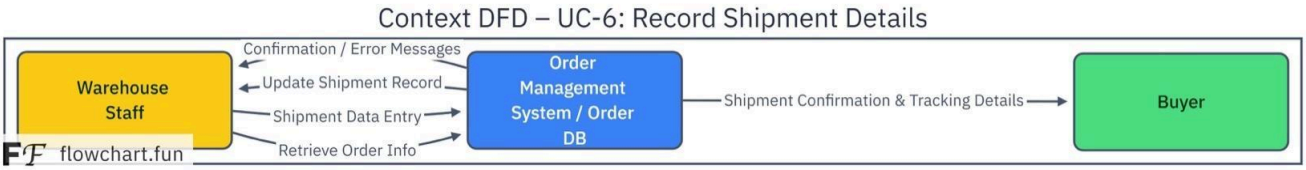
Postconditions:

- Order status = “Shipped.”
- Shipment details are stored and linked to the order.
- Buyer can view shipment details and tracking number in their order history.
- Warehouse logs include the shipment record for auditing.

Exceptions:

- **E1:** Missing required field → System prompts to complete missing data before saving.
- **E2:** Invalid tracking number format → System highlights error and requests correction.
- **E3:** System connectivity issue → System saves entry locally for later synchronization or notifies staff to retry.
- **E4:** Unauthorized access → System denies entry and displays “Insufficient permissions.”

DFD (version 1)



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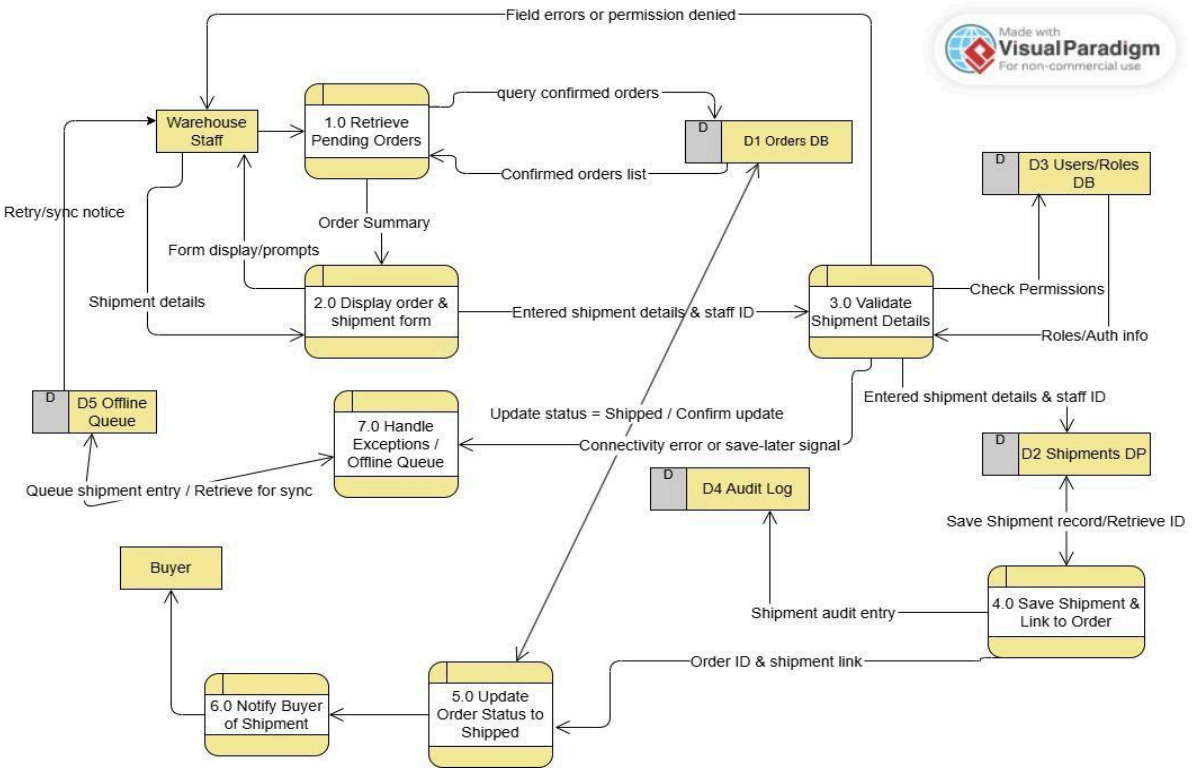
DFD (version 2)



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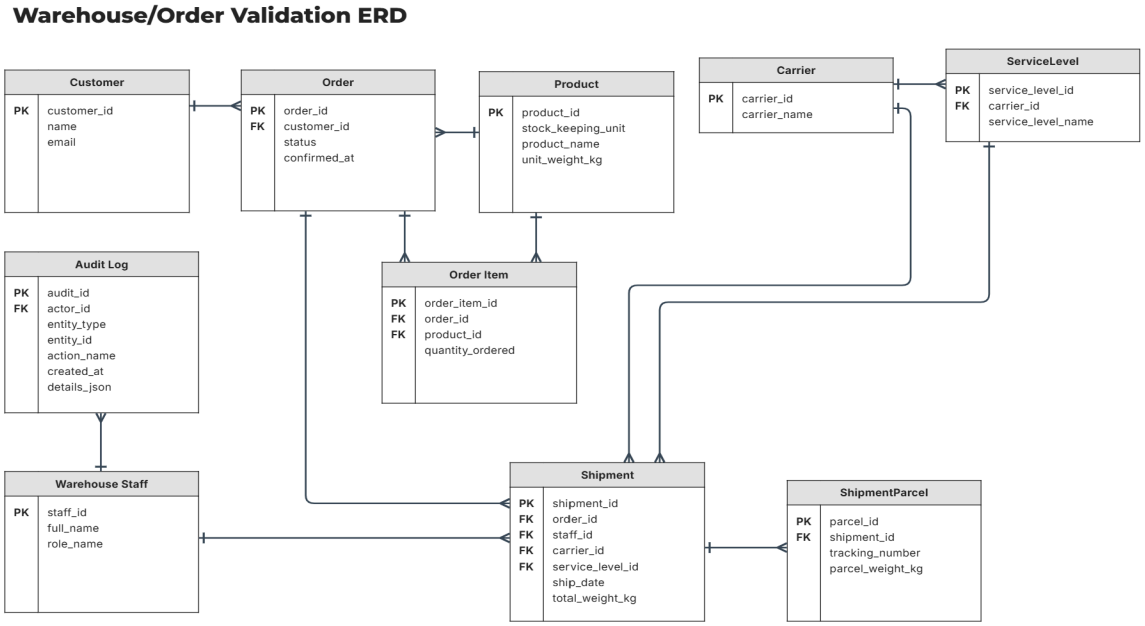
Kelsen Alloway

ERD



Austyn Chan
Faye Liu
Fara Halterman
Finn Wylie
Kelsen Alloway

Use Case Name: Export Customer Data (Faye)

ID: UC-4

Priority: High

Actor: Marketing Manager (primary user), marketing clerk, User

Description: The Marketing Manager exports a filtered set of customer purchase history and preferences for analysis or reporting purposes. This use case covers data filtering based on specific criteria (e.g., date range, product category, customer segment), previewing the results, generating the export file (e.g., CSV or Excel), and downloading the final dataset for use in external tools.

Trigger

- User clicks the “Export” button on the customer data dashboard.

Preconditions

- Marketing Manager is logged into the system.
- Customer purchase history and preference data exist in the database.
- Dashboard is loaded and displaying export option.

Normal Course

- User navigates to the customer data dashboard.
- Selects filters (date range, product category, customer segment) to refine data.
- Clicks the “Export” button.
- System prompts the users to choose a file format (CSV or Excel).
- User selects the desired format.
- User clicks Confirm.
- System generates the file based on selected filters.
- File is downloaded to the User’s device.
- System displays a confirmation message indicating the export was successful.

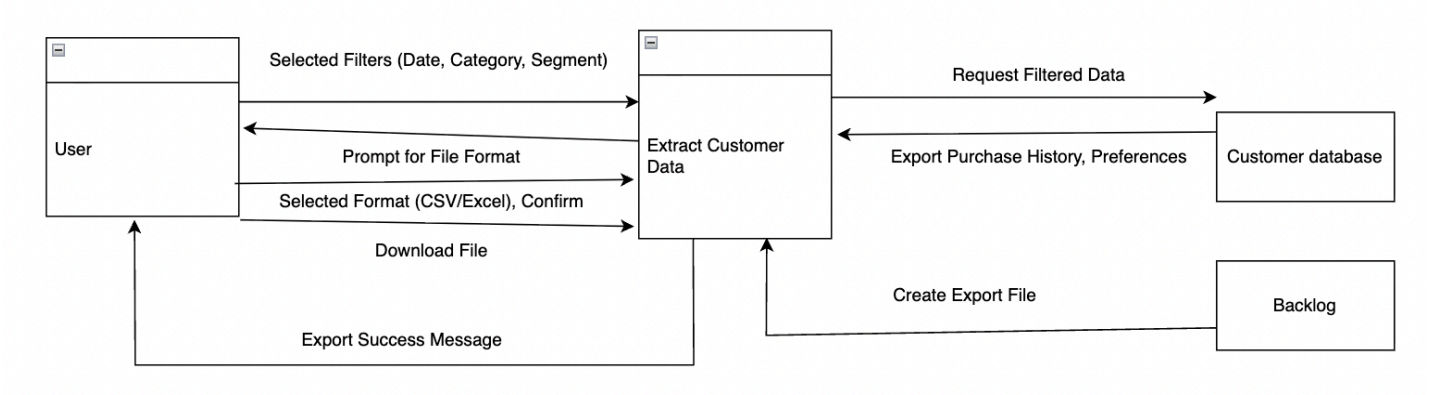
Postconditions

- The filtered customer data is exported and available on the User's device.
- Dashboard remains unchanged and ready for additional exports.
- Filter bar resets to blank after the download, ready for a new selection.

Exceptions

1. **E1:** System displays a message if filters result in no matching records.
2. **E2:** System shows an error if the file cannot be generated or downloaded.
3. **E3:** System displays an error and prompts the user to check their connection if the export cannot be completed due to connectivity issues.

DFD -level 1(version 1)



Austyn Chan
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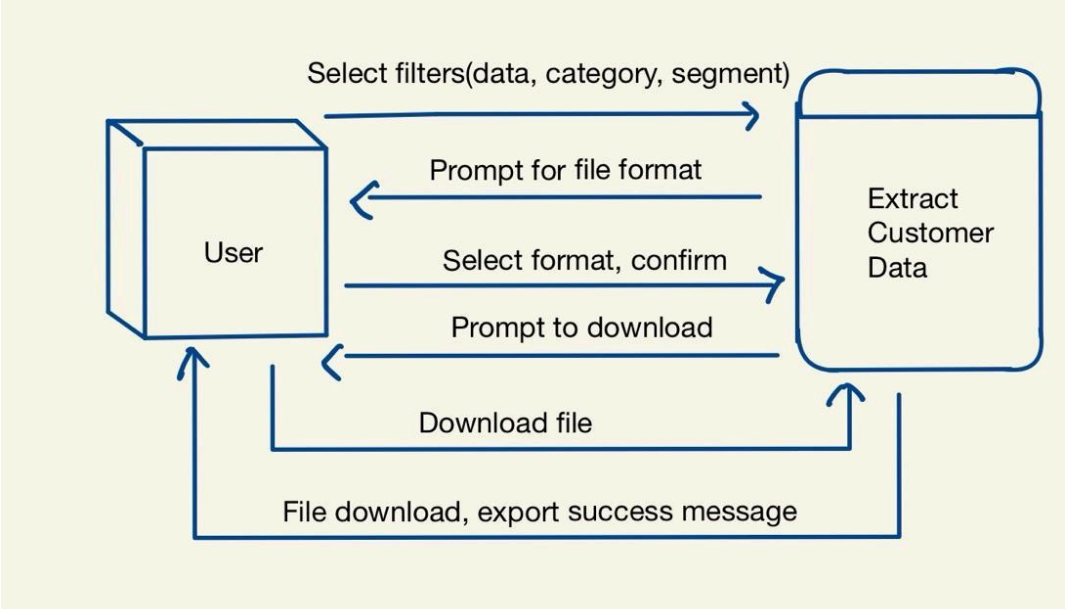
Finn Wylie
Kelsen Alloway

DFD- level 0 (Version 2)



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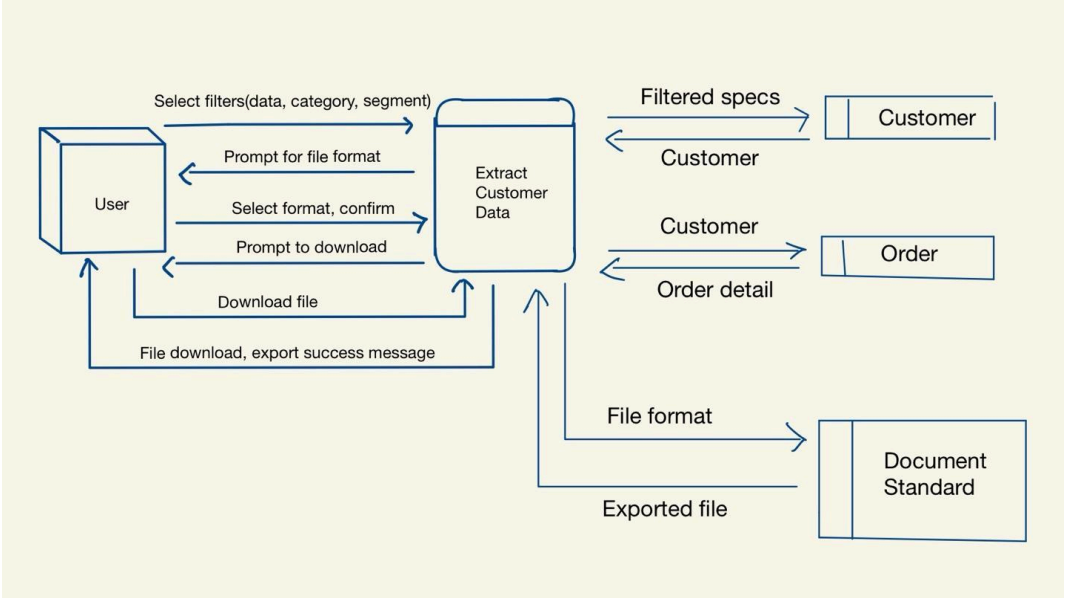

Austyn Chan
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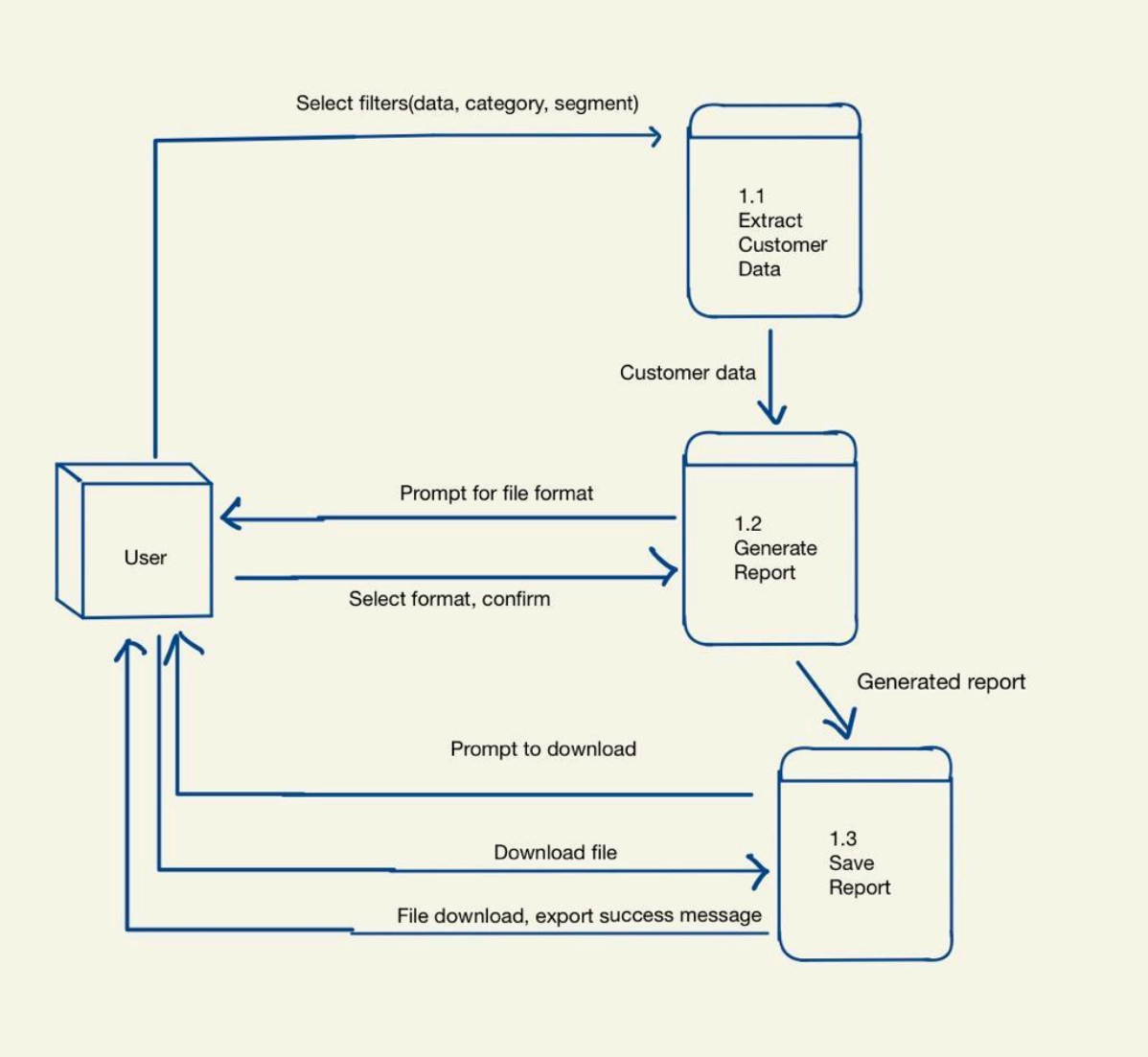
Finn Wylie
Kelsen Alloway

DFD- level 0(Version 3)



Austyn Chan
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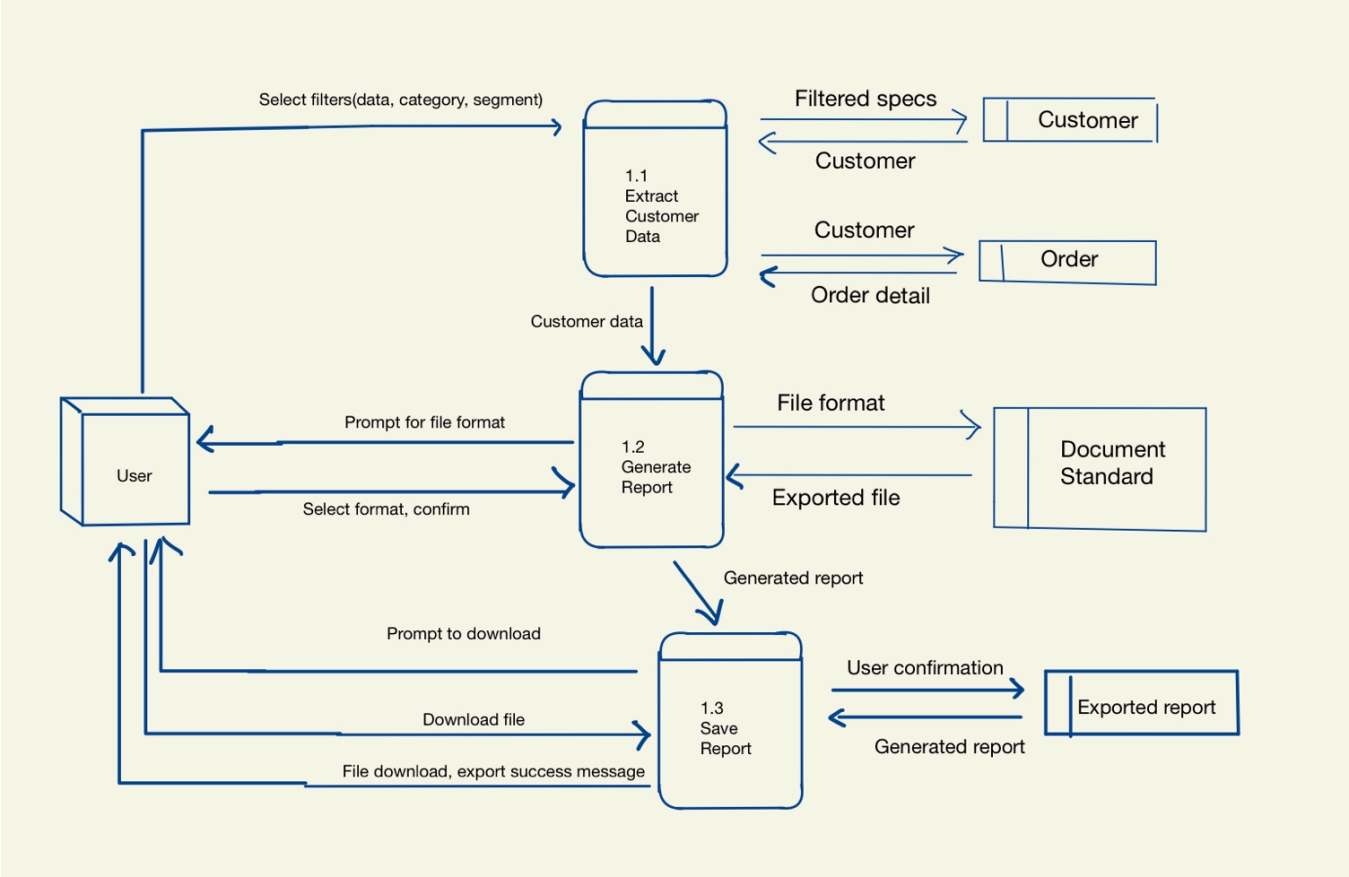
DFD- level 1(Version 3)



Austyn Chan
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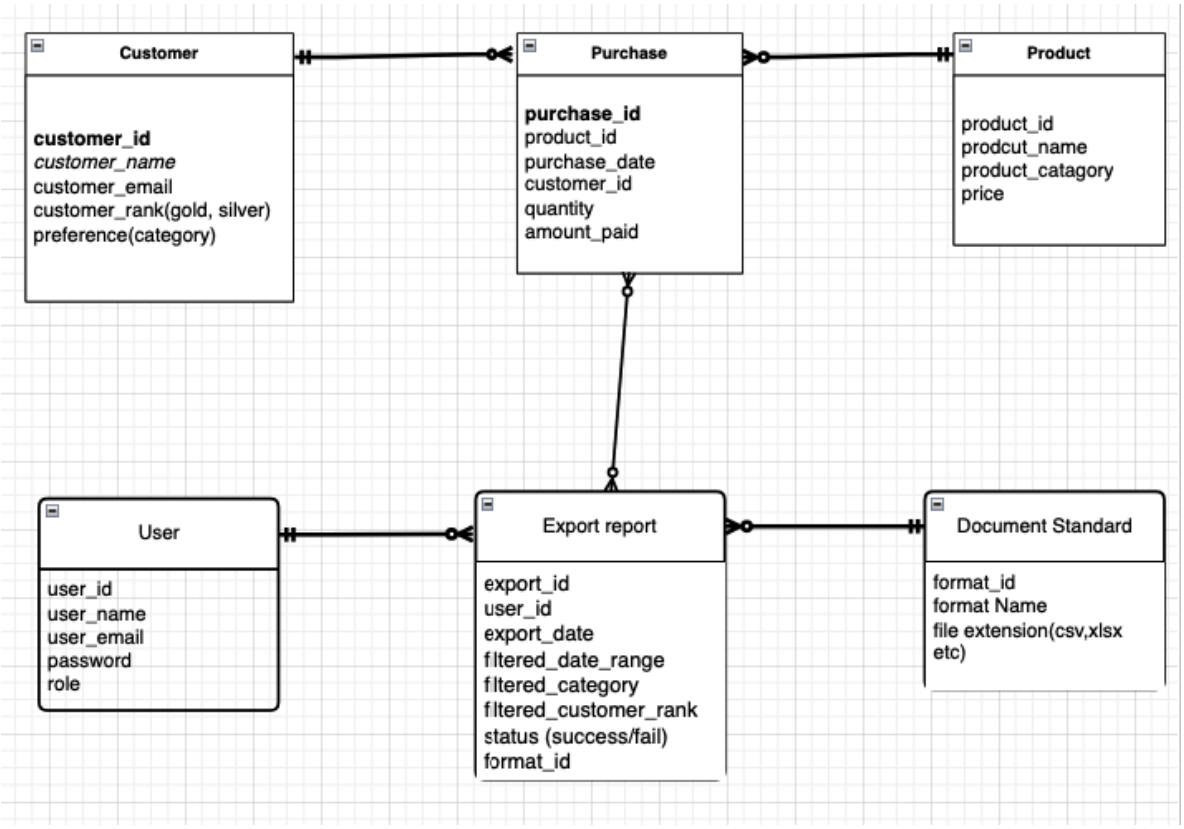

Austyn Chan
Faye Liu
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Finn Wylie
Kelsen Alloway

User Case Name: Generate Sales Summary Report (Austyn)

ID: UC-10

Priority: High

Actor: Manager

Description: The Manager requests the system to generate a Sales Summary report displaying order counts, total revenue, and average sales per day within a specified date range. This report helps management monitor sales performance and revenue trends.

Trigger

- Manager selects “Sales Summary” from the Reports menu and inputs a date range.

Preconditions

- The system contains order and payment data within the selected date range.
- The manager is authenticated with reporting permissions.

Normal Course

- System displays the report criteria form (date range, filters).
- Manager enters the date range and optional filters (e.g., product category, region).
- System validates inputs.
- System retrieves all confirmed/authorized orders within the date range.
- System calculates totals: number of orders, total revenue, and average order value. System generates a tabular and graphical summary (e.g., daily totals, charts).
- The system displays the report on screen.
- The manager may choose to export the report.

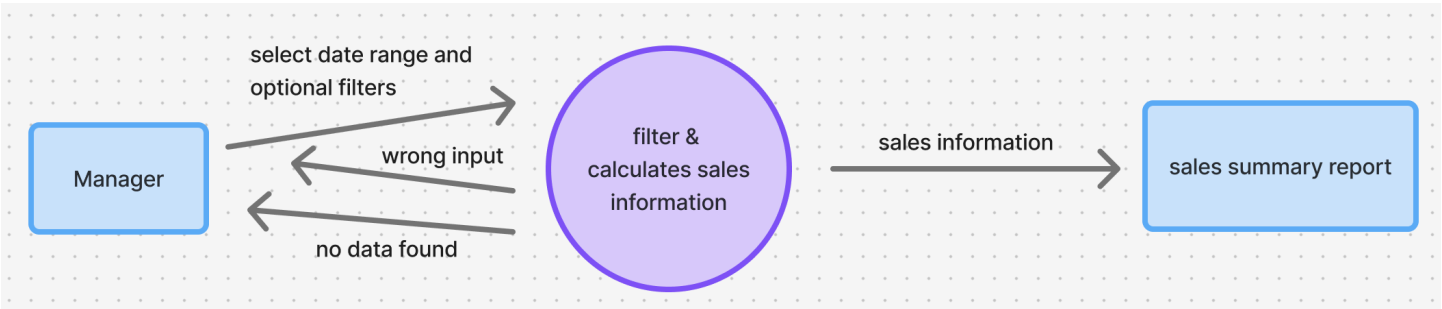
Postconditions

- Reports are generated and viewable.
- Export file (if chosen) is available for download.

Exceptions

- **E1:** Invalid date range (e.g., start > end) → System prompts correction.
- **E2:** No data found → System displays “No orders found for the selected period.”
- **E3:** Database connection error → System shows error messages and logs to the issue.

DFD Level 1 (Version 1)



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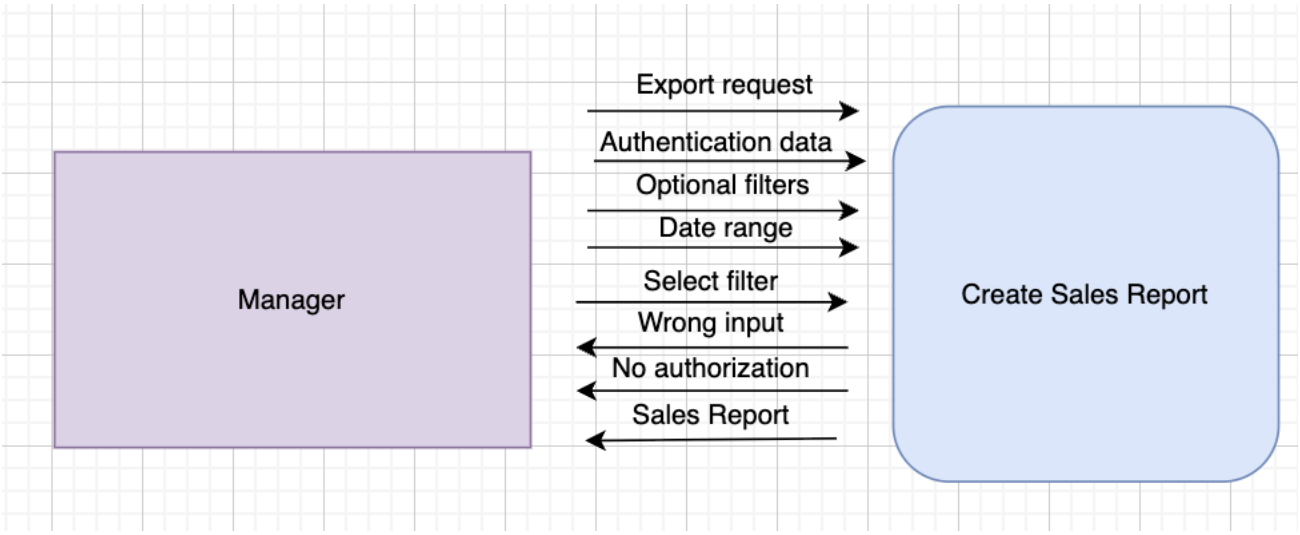

Austyn Chan
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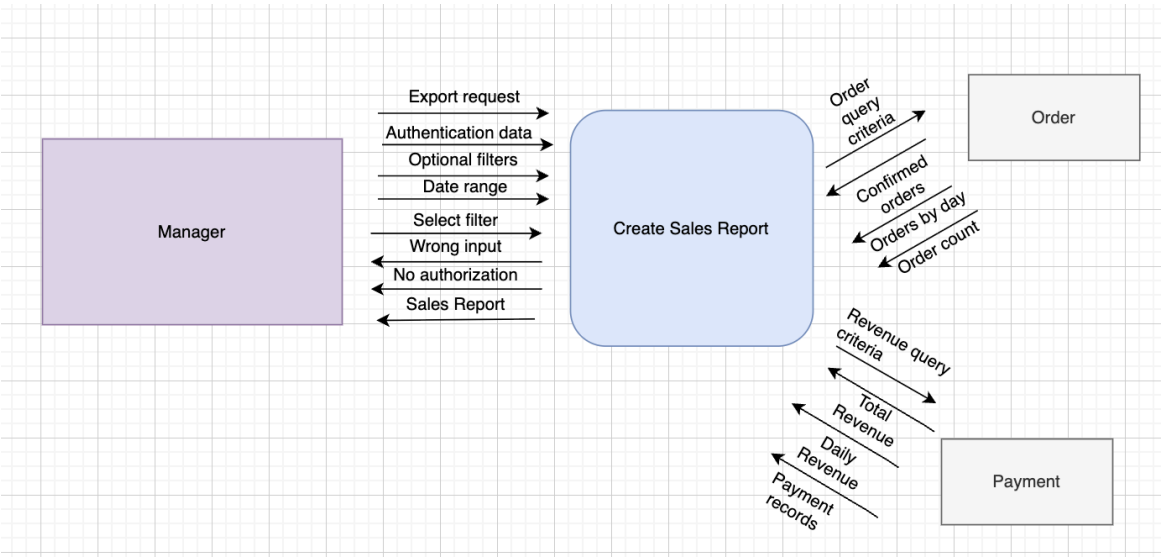

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Austyn Chan
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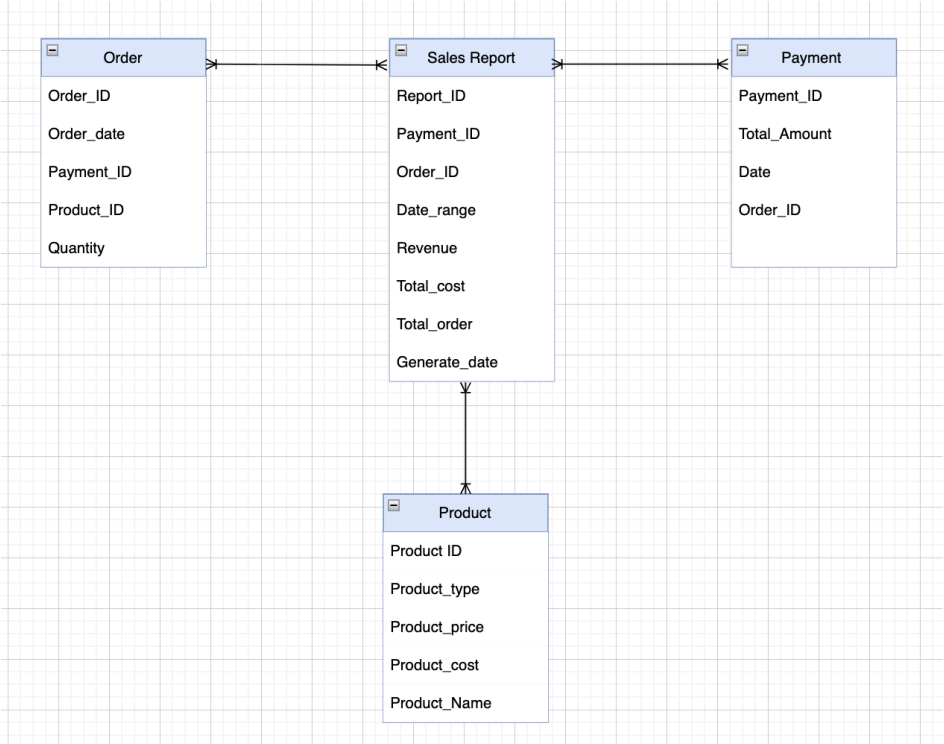

Austyn Chan
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Finn Wylie
Kelsen Alloway

Use Case Name: The system shall store delivery addresses in a consistent format. (Finn)

ID: UC-14

Priority: High

Actor: The system

Description: The user inputs personal delivery information of a customer into the system. The system then stores the data as a JSON or CSV file. This use case covers address storage, and formatting.

Trigger:

User clicks submit on a delivery address entry field.

Precondition:

- Adress is real
- Adress is connected to personal user information
- User has selected a person to input the address to

Normal Course:

- System finds CSV/JSON of associated personal data
- System writes the address in the appropriate fields of the file
- System saves and closes the file
- System provides feedback that the address has been saved

Post Conditions:

- The address is visible in the software with the rest of the user's data
- The address is exportable in multiple formats (email, mail, etc.)
- The address is stored in a standardly formatted document

Exceptions:

- E1: no linked user > System shows decline reason; user can retry inputting address that is linked.
- E2: Address invalid > System prompts corrections or suggests a normalized address.
- E3: File storage full > System alerts that the address could not be saved and prompts why.
- E4: Gateway error/time-out > System shows a non-technical error and allows safe retry.

Austyn Chan
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Finn Wylie
Kelsen Alloway

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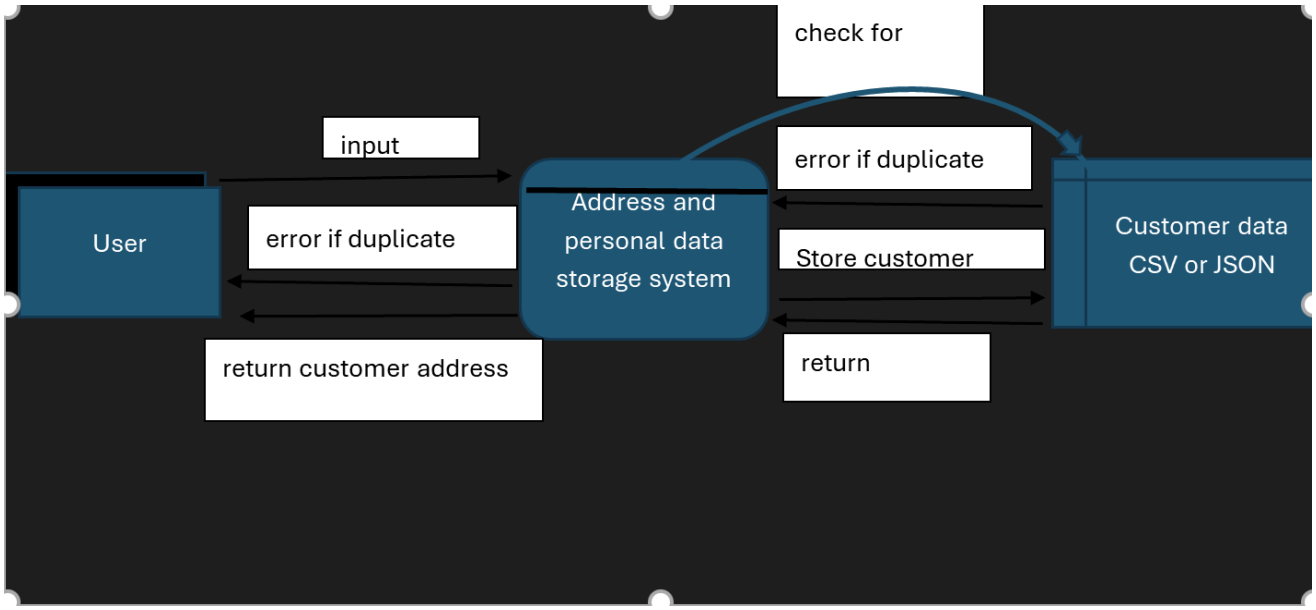

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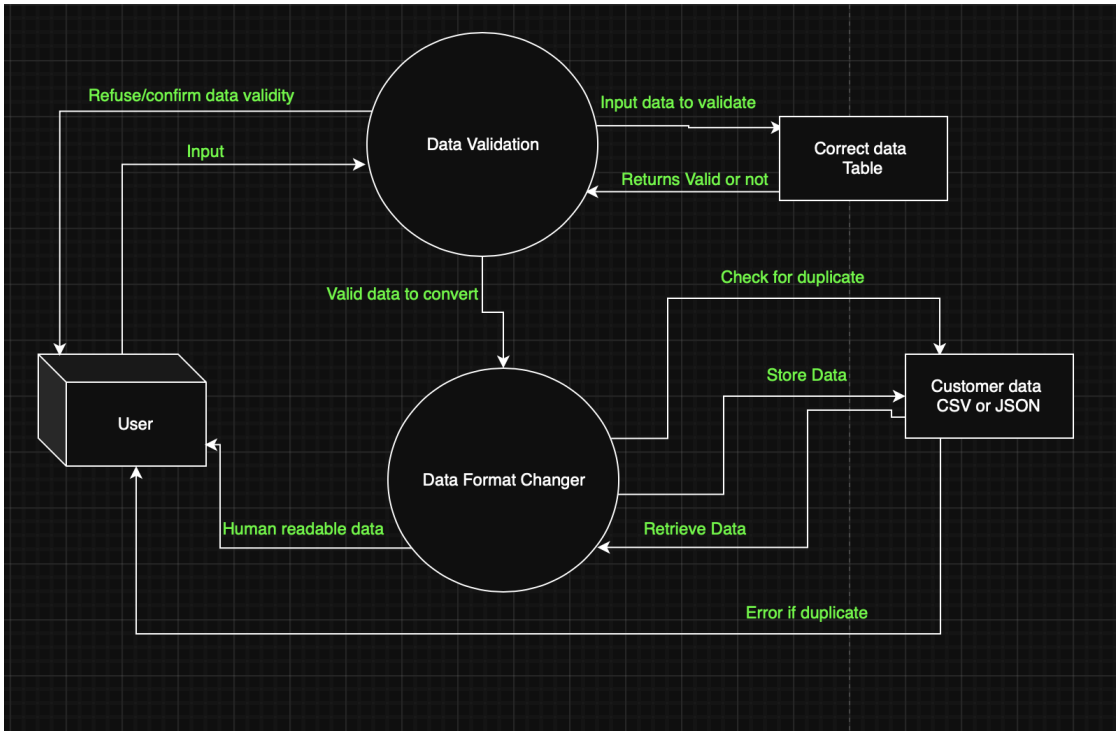

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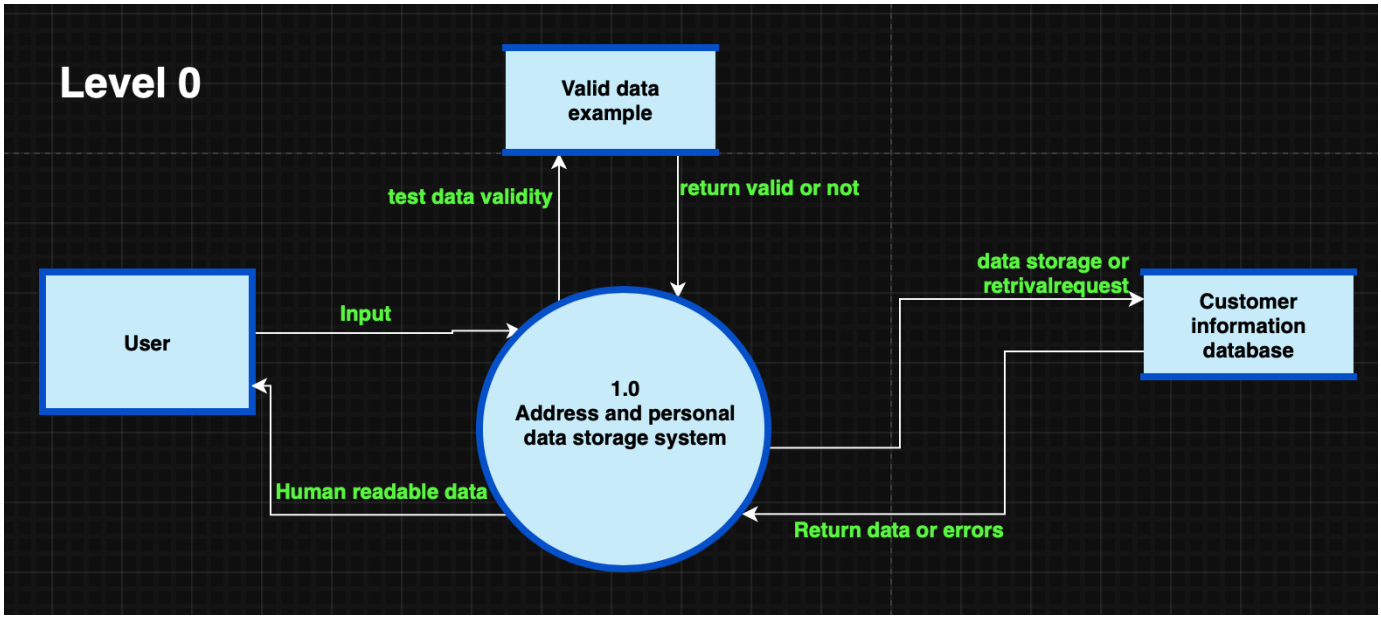

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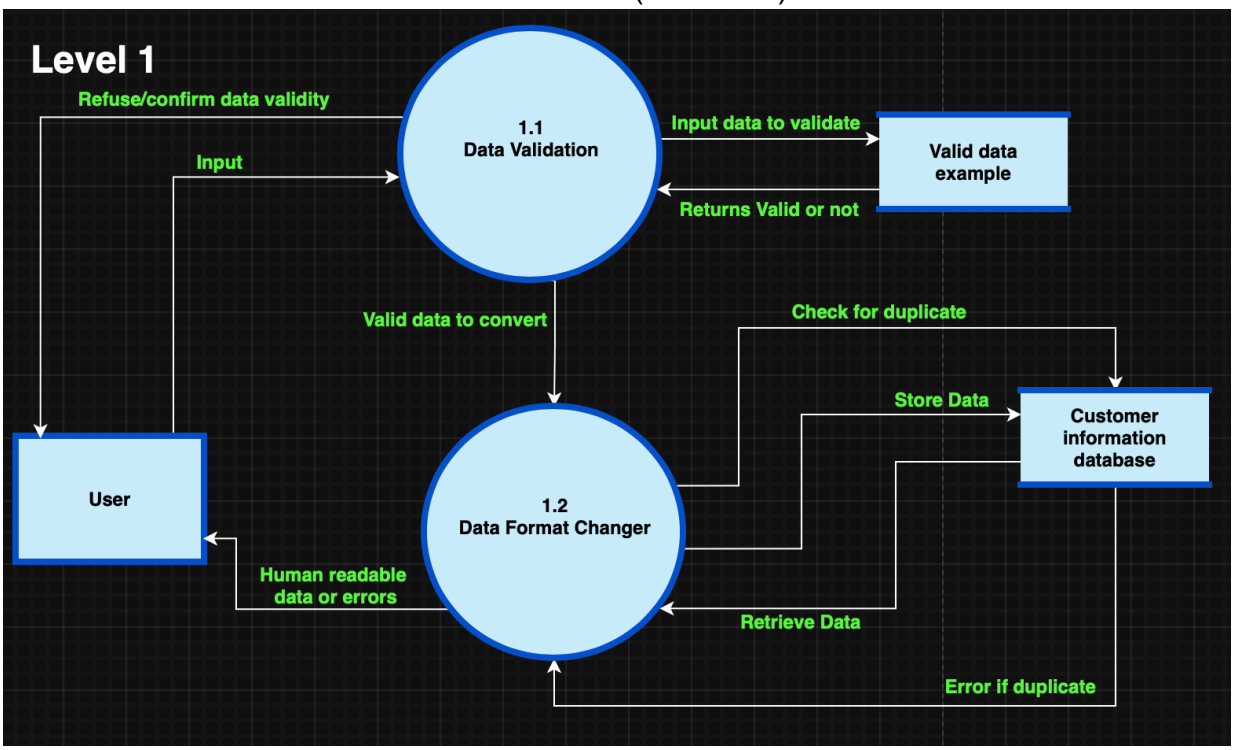

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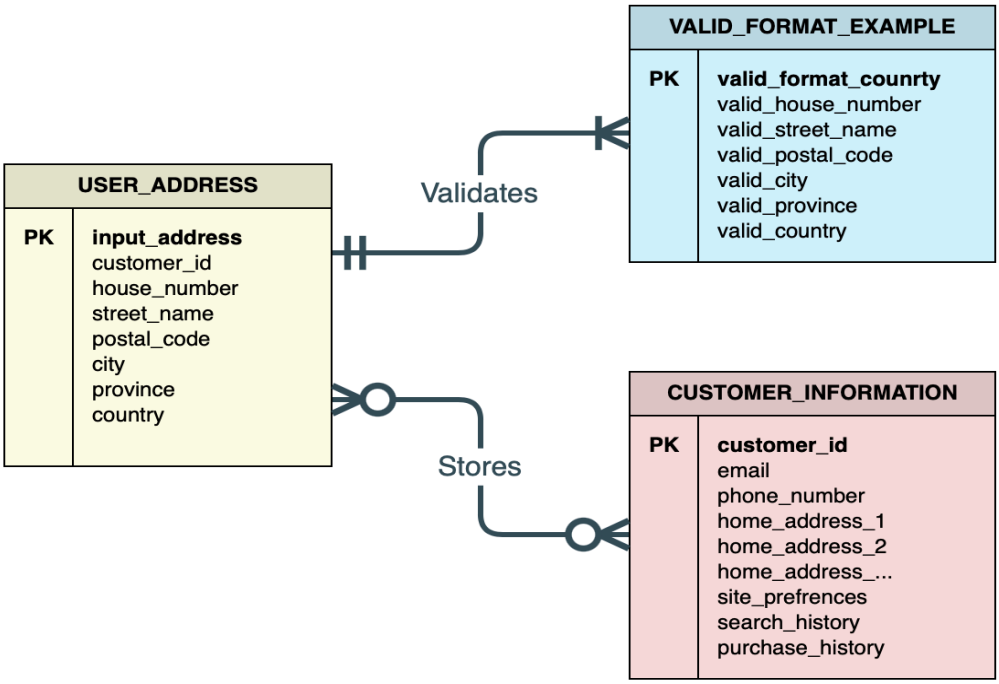


Use Case Summary (Fara)

- **Use Case Name:** Store customer email and phone number
- **ID:** R-12
- **Priority:** High
- **Description:** Clerical staff inputs customer name, email, and phone number into the computer. This use case covers data input.

Actor

- Clerical staff

Trigger

- Clerical staff saves the data to the system.

Preconditions

- Clerical staff is logged into the system.
- Customer name, phone number, and email address are known.

Normal Course

- Clerical staff has customer name, phone number, and email address available on paper.
- Clerical staff navigate to customer data dashboard.
- Clerical staff click “Enter Customer information”
- Clerical staff records customer name, phone number, and email address in the appropriate fields.
- Clerical staff clicks “Save”.
- The system checks if the phone number and email are valid.
- The system prompts the user to confirm the information.
- Clerical staff clicks “Confirm”, or presses Enter on the keyboard.
- The system saves the information if it has been formatted correctly.
- The system displays a confirmation message on success.

Postconditions

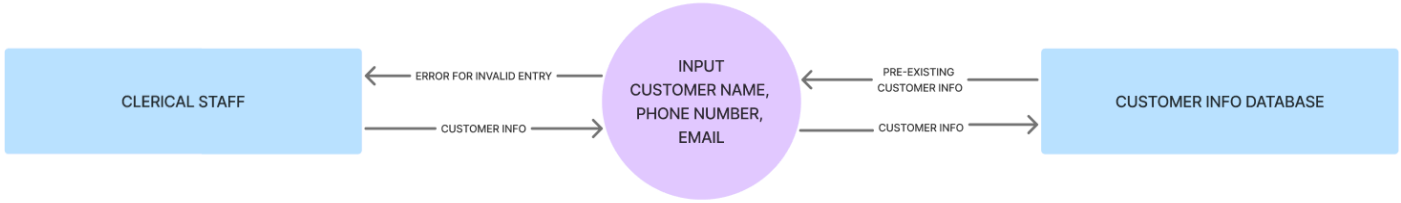
- Customer data is added to the customer database.

- Clerical staff remains on customer data dashboard.

Exceptions

- **Invalid phone number/email:** System prompts user to correct incorrect information; asks if the user would like to continue with missing information (in the case that the clerk was supplied with invalid information)
- **Phone number/email already in the database:** System alerts user the phone number or email is already under the system; system displays the pre-existing customer information and asks if user wants to merge, duplicate, or discard the new input.

DFD - level -1: R-12 (Version 1)



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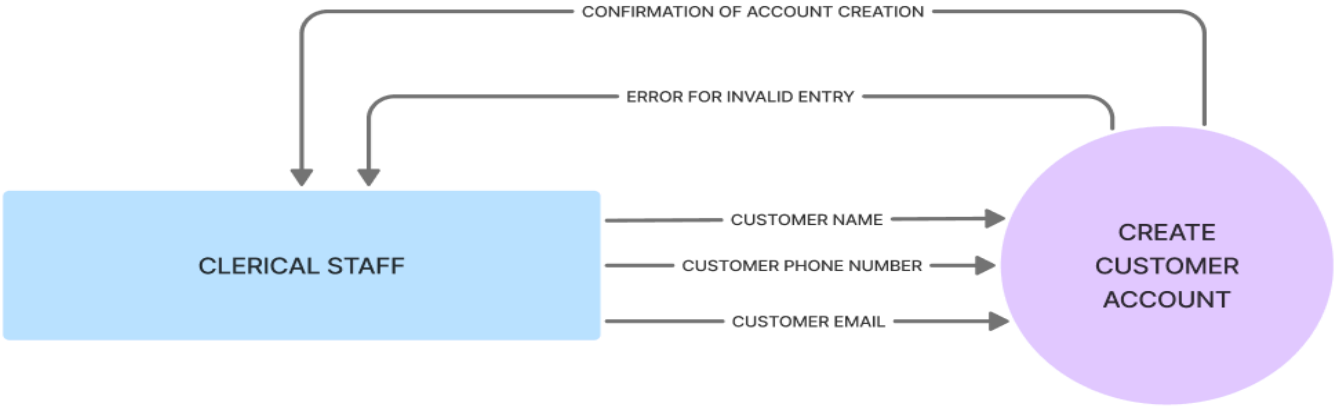

Austyn Chan
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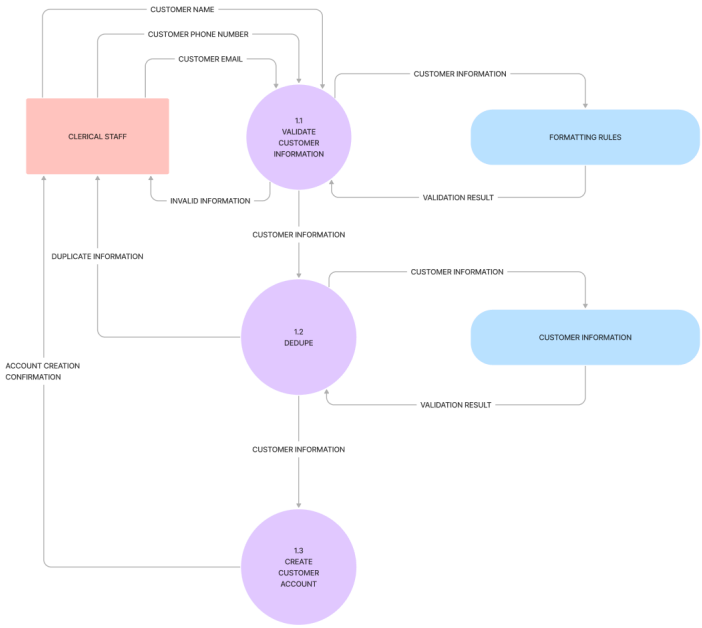
Kelsen Alloway

DFD – level -1: R-12 (Version 2)



Austyn Chan
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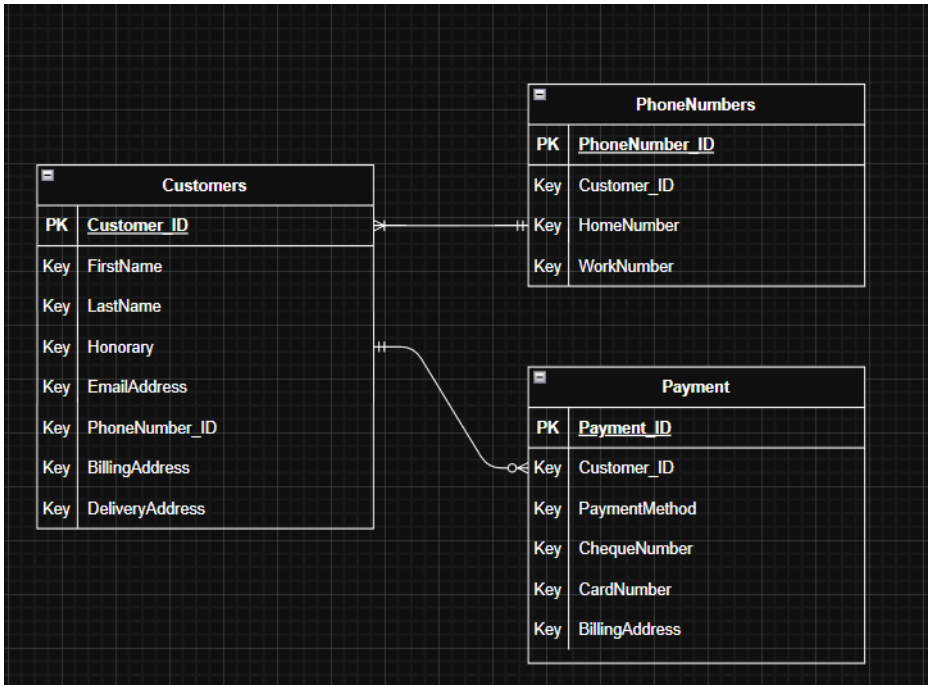

Austyn Chan
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Finn Wylie
Kelsen Alloway

Decision Matrix (Lab 8)

Decision Making Criteria

Decision Making Criteria				
#	Criteria	Why it matters	Definition / how you'll judge it	Weight (100% total)
C1	Phase-1 Functional Requirement Fit	Phase 1 must deliver the core capabilities required by daily operations: accurate stock movement tracking, fast order enquiries with tracking information, clean customer data, and correct financial batch totals. If these essential features do not fit natively, the team cannot meet the business need, operational pain points will remain, and Phase-1 will fail to demonstrate value to leadership.	The system must natively support all Phase-1 required functions (order enquiry, dispatch capture, stock movement, payment batch totals, and basic data-quality controls). Fit will be judged by whether each requirement is available out-of-the-box with no major customization required.	25%
C2	Operatibility & Resiliency	This ensures the system runs reliably day-to-day, even during high workloads or expansion, minimizing downtime and preventing disruptions to orders, dispatch, and customer service.	We will evaluate the vendor's overall operational resilience by looking for safeguards such as daily backups with a 30-day recovery window, automated monitoring and alerts to catch issues early, and a deployment process that avoids releasing updates during busy or sensitive times (e.g., Fridays, month-end), when downtime would have the greatest impact.	15%
C3	Vendor Viability & Roadmap/Scalability	Vendor needs to be able to support store as-is (small scale), and into the future (large, multi country). Must be a stable and trustworthy platform to decrease finacial risk. Must be able to support analytics from multiple internal and external sources. Prices must be realistic for a smaller company who may not use all features.	Work for small and large scale buisinesses. Stable and trustworthy platform. Solid dashboard that works with external sources. Customizable pricing scheme.	15%
C4	Data-Quality Controls	Data quality issues were established as a business need on the System Request	Ability to avoid duplicated user data; linking multiple users to the same address; rules regarding address format and other format sensitive fields	25%
C5	Mobile Performance & UX	Strong mobile performance and a well-designed user experience are essential because both customers and staff increasingly rely on phones and tablets to interact with the system. Fast, intuitive mobile workflows reduce friction, support quicker decision-making, and create a consistent sense of reliability across all touchpoints.	We will assess each vendor on how smoothly their platform performs on mobile devices, how clear and accessible their interfaces are, and how effectively the experience supports key tasks without unnecessary steps or delays.	20%
			Total Weight	**100%**

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Kelsen Alloway

Alternative Matrix

Alternative Matrix

Option 1: Shopify

#	Criteria	Member 1 (Fara)	Member 2 (Finn)	Member 3 (Kelsen)	Member 4 (Austyn)	Member 5(Faye)	Average Score
C1	Phase-1 Functional Requirement Fit	4	4	4	4	5	1.05
C2	Operatibility & Resiliency	4	4	5	4	4	0.63
C3	Vendor Viability & Roadmap/Scalability	5	4	5	5	5	0.72
C4	Data-Quality Controls	5	3	3	3	4	0.9
C5	Mobile Performance & UX	4	4	5	4	4	0.84
						TOTAL	4.14

Option 2: Microsoft Dynamics 365

#	Criteria	Member 1 (Fara)	Member 2 (Finn)	Member 3 (Kelsen)	Member 4 (Austyn)	Member 5 (Faye)	Average Score
C1	Phase-1 Functional Requirement Fit	3	3	5	3	4	0.9
C2	Operatibility & Resiliency	4	5	4	4	5	0.66
C3	Vendor Viability & Roadmap/Scalability	4	5	5	5	5	0.72
C4	Data-Quality Controls	5	5	5	5	5	1.25
C5	Mobile Performance & UX	2	3	3.5	2	3	0.54
						TOTAL	4.07

Option 3: Squarespace

#	Criteria	Member 1 (Fara)	Member 2 (Finn)	Member 3 (Kelsen)	Member 4 (Austyn)	Member 5 (Faye)	Average Score
C1	Phase-1 Functional Requirement Fit	3	4	2	2	3	0.7
C2	Operatibility & Resiliency	5	5	4	5	4	0.69
C3	Vendor Viability & Roadmap/Scalability	3	4	3	3	3	0.48
C4	Data-Quality Controls	2	4	2	2	3	0.65
C5	Mobile Performance & UX	4	4	4	4	4	0.8
						TOTAL	3.32

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Finn Wylie
Kelsen Alloway

COMP 1712 – Set F – Group 22

System Architecture Options (Lab 9)

To: Ivor Merick, Managing Director
From: Group 22, System Analytics
Subject: Architecture Options for Target Enterprises Phase-1 Order Management System
Date: 11/21/25

Introduction

The main objective for Phase-1 is to implement a reliable order management system. This is to replace the current manual workflow that struggles with accuracy, speed, and scalability. The goal of this new system is to reduce duplicates of customer records, speed up order intake, improve dispatch accuracy, and to provide Finance with more stable reporting. This document outlines two possible architecture options, with one as a recommendation. The architectures we have chosen to analyze are Web-based thin client, and Local server with thick client.

Option 1: Web-Based Thin Client (Cloud-Hosted Application)

How it works

Staff access the system through a web browser on desktop or mobile devices, while all business logic such as order enquiries, customer lookup, inventory updates and payment batch summaries runs on the cloud-hosted application servers. Customer, order, dispatch, and stock-movement data is stored in a secure cloud database managed by the service provider. Optional integrations, such as accounting tools, label printing, and analytics dashboards, are supported through standard APIs or plug-in modules. System updates, backups, and security patches are handled automatically by the vendor, reducing operational workload for Target Enterprises.

Main components

- **Client:** Browser (PC / tablet / mobile)
- **Application Layer:** Cloud-hosted application
- **Database:** Cloud database managed by the service provider
- **Network:** Internet connection required
- **Integration Layer:** APIs or plug-in modules

Pros

- **Fast Phase-1 delivery**
  Supports order enquiry, tracking, stock movement, and dispatch entry with minimal setup or custom development.

- **Low maintenance**
  No on-premises servers; backups, uptime, and updates are managed by the cloud provider.

- **Strong mobile experience**
  A lightweight, fast web interface suitable for older mobile devices used by support staff.

Cons

- **Dependency on the internet**
  Network outages directly impact order enquiry and dispatch workflows.

- **Limited deep customization**
  Certain Phase-1 requirements (e.g., specialized reports or data-quality rules) may require add-ons instead of native functionality.

- **Data-quality controls are basic**
  Advanced features such as address standardization or duplicate detection likely require additional modules.

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Option 2: Local Server with Thick Client (Desktop Application on LAN)

How it works

Software is installed on a central in-house server. Each workstation runs a desktop app connected to the server over Target's internal network. All processing and data remain on-site.

Main components

- **Client:** Installed Windows applications for order capture, dispatch, and finance
- **Local Server:** Runs business logic and stores data
- **LAN:** Internal network connecting PCs to the server
- **Scheduled Backups:** Managed manually or via scripts
- **Access Controls:** Roles enforced within the application

Pros

- **Works during internet outages:** Fully internal, good if network reliability improves.
- **High control:** Data stored on-site, no cloud subscription.
- **Performance:** Fast response within the LAN for heavy operations.

Cons

- **Poor fit for current environment:** Patchy network near Dispatch makes adoption risky.
- **Hardware limitations:** Increased client-side demand, increasing the required specifications, and cost of client hardware.
- **Higher maintenance:** Requires local server upkeep, patching, and manual backup checks.
- **Limited mobility:** Harder to support order enquiries on phones.
- **Slower rollout:** Installation required on each workstation; harder to update with Finance's timing constraints.

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Kelsen Alloway

Recommendation: Option 1 – Web-Based Thin Client (Cloud-Hosted)

Based on Target's operational realities—shared workstations, patchy local network, reliance on mobile devices for Order Enquiry, and the need for rapid Phase-1 delivery—**a cloud-hosted web application is the best fit**.

This architecture:

- aligns with Ivor's request for fast mobile Order Enquiry and lightweight pages
- supports Operations' need to capture dispatch details cleanly
- protects Finance's month-end workflow by enabling safe, low-impact releases
- reduces data-quality risk through shared, real-time validation
- provides stronger resilience (daily backups, audit trails, restore options), and supports future expansion without major infrastructure investment.

A cloud-based thin-client system offers Target Enterprises the stability, usability, and scalability needed for Phase-1 success and long-term growth.